

Ahmad Sultan Mirzam Ambary

Bekasi, Indonesia | (62) 81210011827 | ahmad.irza18@gmail.com | github.com/AhmadSultanMA/ | web-porto-ahmadsultanma.vercel.app/

A results-driven Information Technology graduate with hands-on experience in building scalable applications across web, mobile, and AI/ML platforms. My expertise includes deploying end-to-end AI pipelines and full-stack development using modern frameworks. I have a proven track record of ranking in the top 100 of the Google Solution Challenge, showcasing my commitment to innovation and technology implementation.

EDUCATION

Brawijaya University

Aug 2022 – Jul 2026

Bachelor's Degree in **Informatics Engineering**

CGPA: 3.83/4.00

Related Coursework: Database, Human-Computer Interaction, Object Oriented Programming, Artificial Intelligent, Machine learning, DevOps, MLOps, Internet of Things, Website Development, Application Development.

Madinatul Quran Vocational High School

Jul 2019 – Jun 2022

Related Coursework: Database, Website Development, Application Development.

Tools: Visual Studio Code, Android Studio, Github, Postman, Xampp, Figma, Supabase, Docker Desktop, Anaconda, Google Colab, Amazon Web Services, NetBeans, Oracle Virtual Box, Firebase, Google Cloud Platform, Vertex AI, Arduino, Antigravity, Prometheus, Grafana.

Languages & Frameworks: Flutter, Dart, TypeScript, Kotlin, React.js, React Native, Java, Golang, Next.js, Vue.js, Python, PHP, SQL, HTML/CSS, C++.

WORK EXPERIENCE

PT. Telkom Indonesia

Jakarta, (On-site)

Field Work Internship | AI Engineer

Jul 2025 – Sep 2025

- Designed and deployed an end-to-end Computer Vision pipeline using YOLOv11n segmentation for real-time Personal Protective Equipment (PPE) compliance monitoring.
- Curated, annotated, and balanced a custom dataset of 14 classes (covering wear/non-wear states), achieving a Mean Average Precision (mAP) of 0.708.
- Integrated the AI inference pipeline with CCTV static IP feeds and Network Video Recorders (NVR) to automate safety compliance alerts in industrial environments.

Bank BTPN Syariah

Jakarta, (Remote)

Project Based Internship | Fullstack Developer

Jan 2024

Participated in a project-based internship, gaining valuable experience in Vue.js and Golang, and contributing to the development of a key API for the final project.

- Completed a structured project-based internship program, acquiring hands-on experience in full-stack web development using Golang and Vue.js.
- Independently developed a final project featuring a Golang-based RESTful API for efficient data handling and integration.
- Successfully completed the project-based internship, earning the qualification of a qualified student, demonstrating both technical and professional competence.

PT. Telkom Indonesia

Bandung, (On-site)

Field Work Internship | Frontend Developer

Aug 2021 – Oct 2021

- Refactored a monolithic codebase of 5,000+ lines of code into modular, reusable components, significantly improving code maintainability and decreasing code redundancy.
- Developed responsive front-end interfaces integrated with the Laravel framework, collaborating with senior student developers to deliver project features on schedule.
- Created and maintained comprehensive technical documentation to ensure clear technical specifications and smooth handover processes for the development team.

PROJECTS

Lecturer's Project

Malang, (On-site)

Signesia | Full-Stack 3D Motion Pipeline Developer

Feb 2026 – Jul 2026

Designed and implemented an end-to-end pipeline that converts raw MediaPipe landmark data into real-time 3D avatar animations for BISINDO (Indonesian Sign Language), spanning both the Python inference backend and the React Three Fiber rendering frontend.

- Architected a hybrid 2D-to-3D lifting pipeline that applies MotionBERT (DSTformer) for body pose reconstruction while using MediaPipe's native world-space landmarks for hand and facial depth, remapping MediaPipe's 33-point pose to Human3.6M 17-joint format and anchoring hand landmarks to body wrist positions via translation offset correction.

- Implemented a quaternion-based bone retargeting system in TypeScript/React Three Fiber that maps lifted 3D pose data onto Avaturn GLB humanoid avatar rigs for smooth, real-time skeleton animation, including facial blendshape drivers for non-manual sign expressions.
- Diagnosed and resolved core retargeting defects including left-hand wrist mirroring, finger hyperextension from axis sign errors, and arm-body mesh penetration by tracing root causes to pixel-space vs. world-space landmark mismatches.

Capstone Project

NEO PARK | Lead AI & IoT Systems Engineer

Malang, (On-site)

Feb 2025 – Jul 2025

Developed an intelligent parking system integrating computer vision and IoT, recognized as the "Best Capstone Project" in the cohort/departement.

- Fine-tuned YOLOv8 to detect cars to manage parking slots with over 90% confidence and built a real-time parking system combining IoT sensors and camera input, achieving 1-second latency.
- Architected the integration layer to seamlessly synchronize YOLOv8 inference results, IoT sensor telemetry, and the frontend display.
- Wrote comprehensive pytest test suite covering REST API endpoints, detection logic, and status handling, using mocking and fixtures to isolate YOLOv8 inference and Prometheus metrics for reliable CI testing.

Faculty of Computer Science

SIMABA 2024 Website | Lead Fullstack Developer & Head of IT Division

Malang, (On-site)

Jun 2024 - Aug 2024

Led the development of a comprehensive website for the PKKMB (SIMABA 2024) event as the Head of the PIT Division.

- Led a cross-functional IT division of 8 developers to design, build, and deploy the official freshman orientation platform, serving over 900 active users.
- Architected and implemented core full-stack features using Laravel and React.js including secure assignment submission, automated attendance tracking, grading portals, and dynamic group distribution algorithms.
- Optimized database queries and front-end performance to handle concurrent traffic from hundreds of students simultaneously with zero downtime and minimal rendering lag.

Google Solution Challenge 2024

UrDENTIST APPS | Mobile Developer

Online

Dec 2023 - Mar 2024

Developed a mobile application using Flutter aimed at improving dental health by detecting dental caries, offering tailored treatment recommendations, and providing daily tasks for maintaining oral hygiene.

- Secured a Top 100 spot globally in the Google Solution Challenge 2024 (competing against thousands of projects) for outstanding innovation in healthcare technology.
- Developed the cross-platform frontend using Flutter and Dart, implementing highly responsive UI layouts, interactive animations, and efficient state management.
- Integrated machine learning APIs to enable automated dental caries detection from user-uploaded images, dynamically rendering personalized treatment recommendations.

Basic Computing Community

Sportix Website | Backend Developer

Malang, (Onsite)

Mar 2023 - Apr 2023

Developed a web-based sports facility and equipment rental platform designed to streamline booking workflows and automate payment verification.

- Engineered the backend architecture for a sports facility and equipment rental booking platform using the Laravel framework, building 35+ RESTful API endpoints within 1 month, and securing admission into the prestigious BCC.
- Integrated the Midtrans payment gateway to automate transaction processing, callback handling, and secure billing workflows.

CERTIFICATION

- AWS Academy Data Engineering – AWS (2025)
- Top 100 Google Solution Challenge – Google (2024)

ACTIVITIES

- Head of PIT Division PKKMB 2024.
- Teaching Assistant for OOP (Java) 2024.
- App Developer at RAION Community 2024.
- BEM FILKOM's Information and Technology Development Staff 2023.
- Staff of Information and Technology Division Brawijaya Tournament 2023.

Language: Bahasa Indonesia (Native), English (BEPT TOEFL Based: 567).